



Instructor: Date:

Learner: Regional Branch:

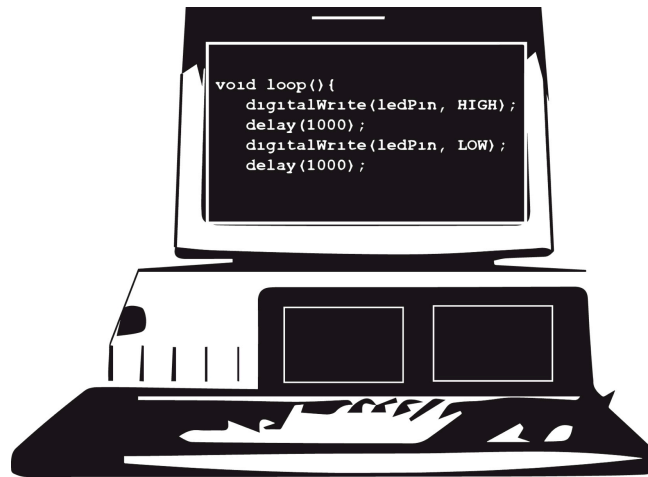
Program: Center:

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**Let's Start!**

Looking into the Origins of Microcomputers

**Learning Outcome:**

Identify information about the history of software and its characteristics in simple spoken and written situations.

Ask your partner:

- What are the advantages of a portable computer?
- Remember the size of the ENIAC. Why do you think the creation of smaller computers was an important change for human kind?

Can Do Statements:

- I can talk about the origins of minicomputers.
- I can talk about the evolution of minicomputers in the industry.
- I can discuss the importance of integrated circuits in the evolution of microcomputers.



Let's Explore!

Let's talk a little bit about the history of microcomputers



Instructions: Read the text below.

Sophia: Were you able to read about microcomputers yesterday?

Jeffery: Yes. Let's review what we read in the article. Why were microcomputers a good idea?

Sophia: Computers were first designed for governments and large corporations. However, some people in the industry thought that computers didn't need to be so exclusive. Computers were very expensive and big, so developers had to find a way to reduce the price and size. So, they had to face the big question: How can we make computers more affordable? Their solution was to replace the vacuum tubes with transistorized models for making smaller and cheaper computers.

Jeffery: Can a solution like this make a real difference?

Sophia: Certainly. The innovative microcomputers brought a new class of users. After that, scientists and engineers were able to afford computers.

Jeffery: How were these new computers innovative?

Sophia: Thanks to the new technology and the economies of scale, the prices of microcomputers decreased and by 1970 there were around 50 companies making microcomputers. These technology advances are still very important today.

Jeffery: Were these minicomputers used only by scientists and engineers?

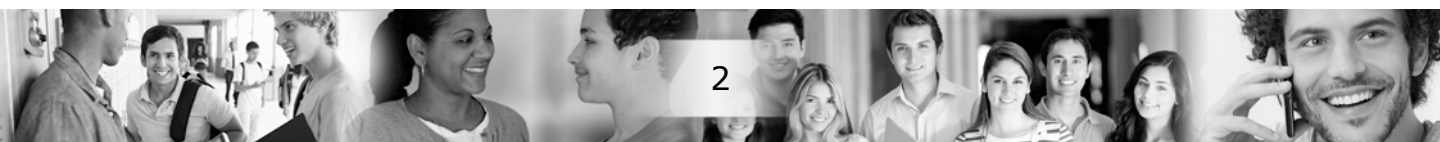
Sophia: At the beginning, they were. However, during the 1980s, computers were small and cheap enough for making them affordable for individuals.

Jeffery: How were they able to create more affordable computers?

Sophia: It was thanks to the invention of the integrated circuit. Many transistors were included in a single chip of silicon, making computers smaller and affordable for individuals.

Jeffery: How can integrated circuits make computers smaller?

Sophia: It's simple. As more transistors fit in a smaller space, computer memory, logic circuits, and other components take less room, and computers become smaller.



**Let's Practice!**

Jeffrey has three important questions about the history of microcomputers.

Instructions: Read the sentences. Write the missing word in the blanks. Use Were the, Why were microcomputers, Were integrated, Were the prices. Use capital letters where necessary.

1. _____ important for scientists and engineers?
2. _____ circuits a technological advance for making computers smaller?
3. _____ prices of microcomputers high by 1970?

**Let's Practice!**

Let's go over* some vocabulary to talk about microcomputers.

*To "go over" means to review something.

Instructions: Organize the mystery sentence.

1. were / the / expensive / microcomputers, / of / invention / computers / big / and /
Before / machines.

2. circuits. / thanks / Microcomputers / smaller / inclusion / to / of / the / were /
integrated

3. fifty / on / working / companies / There / approximately / microcomputers. / were





Extension Activity

The History of Minicomputers

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Make a list of important facts about the origins of minicomputers.
- Take notes of some crucial dates and moments in the evolution of the minicomputers industry.
- Use all the information you have to talk about the microcomputer. Be ready to ask and answer questions.

Self-Assessment

Yes

No

- I list important facts about the origins of minicomputers.
- I recognize some important moments in the evolution of the minicomputers industry.
- I prepare a presentation for my classmates and instructor.

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Target Words

- **economies of scale** (noun): the inverse relationship that occurs when the quantity of a product increases and its price decreases
e.g. Thanks to the new technology and the economies of scale, the prices of microcomputers decreased.
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e.g. Thanks to the new technology and the economies of scale, the prices of microcomputers decreased.
- **room** (noun): synonym for space
e.g. As more transistors fit in a smaller space, computer memory, logic circuits and other components take less room, and computers become smaller.
- **however** (connector): word used to contrast two ideas or statements
e.g. However, some people in the industry thought that computers didn't need to be so exclusive.
- **transistorized** (adjective): something that is equipped or that contains transistors
e.g. Their solution was to replace the vacuum tubes with transistorized models for making smaller and cheaper computers.
- **afford** (verb): to have enough money or capital to buy something
e.g. After that, scientists and engineers were able to afford computers.



Learning Points

Simple Past Tense of “verb be”, “were” in questions

We can use the **simple past tense** of the **verb be** for asking questions about the state of things or people in the past.

When we want to use the simple past tense of the verb **be** for asking questions, we use the following structure:

Verb Be in Simple Past (Was/Were) + Subject + Complement + Question mark (?)

Example: **Were microcomputers** *an important invention?*

Remember that when we ask a question, we place the verb before the subject.

When we have an information question, we place the interrogative adverb **before the verb be**. Let's consider the following example:

Example: **How were** *the first microcomputers?*

Examples

- *Why were microcomputers a good idea?*
- *How were these new computers innovative?*
- *Were these minicomputers used only by scientists and engineers?*



Learning Points

Verb Can in Questions

You use the modal verb **can** to talk about possibility, ability, requests or permission. Here, we'll focus on the use of this modal verb in questions.

Take a look at the following question:

Can you **read** that paragraph about microcomputers?

Notice that the structure of this question is similar to the one explained in the previous Learning Point. You place the verb **can** before the subject and then, you place the verb.

It's important to say that the verb *can* remains the same regardless of the subject of the sentence. Besides, the main verb of the sentence stays in its base form. As you can see, the verb in the example is **read** and it takes the base form.

Now, in terms of meaning, notice that in the question we use the modal verb **can** to indicate **a request or an instruction**.

Take a look at the following examples:

Can you **help** me understand the functioning of microcomputers? (Request)

Can I **read** your text about ENIAC? (Request)

It is also possible to use **information questions** when we use the verb **can** in interrogative sentences, just follow the structure below:

Interrogative adverb (What, How, Where, Why, etc.) + **Modal verb can** + Subject + Complement + Question mark (?)

Example: **How can** microcomputers change the economy? (Ability)

Examples

- *How can we make computers more affordable?*
- *How can integrated circuits make computers smaller?*
- *Can a solution like this make a real difference?*

